

AN ADDITIVE OVERLAPPING DOMAIN DECOMPOSITION METHOD FOR THE HELMHOLTZ EQUATION*

WEI LENG[†] AND LILI JU[‡]

Abstract. In this paper, we propose and analyze an additive domain decomposition method (DDM) for solving the Helmholtz equation with the Sommerfeld radiation condition. In the proposed method, the computational domain is partitioned into structured subdomains along all spatial directions, and each of the subdomains contains an overlapping region for source transferring through the perfectly matched layer (PML) technique. At each step all subdomain PML problems are solved concurrently, and then the horizontal, vertical, and corner directional residuals on each subdomain are passed to their corresponding neighbor subdomains as the sources for the next step. This DDM is efficient in nature and theoretically shown to produce the exact solution for the PML problem with constant medium defined in the whole two-dimensional space. A slightly modified version of the method for bounded truncated domains is next developed for its use in practice and an error estimate is rigorously proved. Various numerical experiments in two and three dimensions are conducted on the supercomputer Tianhe-2 Cluster to verify the theoretical results and demonstrate excellent performance of the proposed method as an iterative solver or a preconditioner.

Key words. additive domain decomposition method, Helmholtz equation, perfectly matched layer, truncated domain problem, error analysis, parallel computing

AMS subject classifications. 65N55, 65F10, 65F08, 65Y05, 65Y20

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1. Introduction. In this paper, we consider the well-known Helmholtz equation defined as follows:

$$(1.1) \quad \Delta u + k^2 u = f \quad \text{in } \mathbb{R}^n$$

with the Sommerfeld radiation condition

$$(1.2) \quad r^{\frac{n-1}{2}} \left(\frac{\partial u}{\partial r} - iku \right) \rightarrow 0, \quad \text{as } r = |\mathbf{x}| \rightarrow \infty,$$

where $u(\mathbf{x})$ is the unknown function, $f(\mathbf{x})$ is the source, $k(\mathbf{x})$ denotes the wave number, and $k(\mathbf{x}) = \omega/c(\mathbf{x})$ with ω being the angular frequency and $c(\mathbf{x})$ the wave speed. Among many discretization and solution techniques for solving the Helmholtz equation (1.1) with the condition (1.2), the domain decomposition method (DDM) [33, 27] is a very effective approach when the problem scale is very large and has been thoroughly studied for several decades. The basic idea is as follows: truncated with an absorbing boundary condition, all subdomain problem are (exactly or approximately) solved during each step, and the subdomain solutions are then passed to their respec-

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tive neighbor subdomains via interface conditions to carry on the wave propagation process. The DDM is also especially popular to be used as a preconditioner to the iterative solver for the global discrete system.

The DDM was first used to solve the Helmholtz problem by Després in [10], and then various DDM algorithms have been developed based on the choices of the transmission condition, which mostly relates to the approximations of the Dirichlet to Neumann (DtN) map. A nonlocal transmission condition was first introduced by Collino, Ghanemi, and Joly [6], where a square-root type operator is used. Gander and co-authors introduced optimal nonlocal interface conditions [16, 18, 17, 20, 21, 19], where the best approximations of zero and second orders are used in the transmission conditions. Douglas and Meade also introduced a second order transmission condition [11]. Boubendir, Antoine, and Geuzaine [4] proposed a nonoverlapping Schwarz method using the Padé-localized square root operator for the approximation of the DtN map, and the effective convergence is quasi-optimal. Employing the perfectly matched layer (PML) in the transmission conditions was first proposed in [32, 36].

The sequential sweeping DDM preconditioner first proposed by Engquist and Ying [14, 15] is shown to be quite effective and followed by the development of various sweeping type DDMs, which differ mainly in the transmission condition at the interfaces of subdomains. The transmission condition of Engquist and Ying [14, 15] is that the residual in just one layer is taken as the source. The DDM introduced by Stolk [34] uses a transmission condition that the derivatives of the solution are treated as a delta source. The double sweeping preconditioner developed by Vion and Geuzaine [38] uses a mixed boundary condition that involves the DtN map as the transmission condition. The polarized trace method developed by Zepeda-Núñez and Demanet [40] imposes a transmission condition that both single and double layer potentials are taken as delta sources. Instead of using the delta source approach, the source transfer DDM (STDDM) [7, 8] proposed by Chen and Xiang uses a smooth source coming from the residual as the transmission condition, and furthermore, an error estimate of the method has been rigorously proved. Recently, Gander and Zhang [23] established a common formulation for the sweeping preconditioners [14, 15], source transfer [7, 8], single layer potentials [34], and polarized traces [40] in the context of optimized Schwarz methods. It is also found in [23] that the idea of source transfer is related to the deferred correction. It is noted that the sweeping type DDMs often partition the domain in one direction, and the sweeping order would be along the direction forward and backward. Two orthogonal directions sweeping could be done in a recursive way, for instance, as proposed by Liu [30] using the sweeping preconditioner [14, 15], and by Wu [12] using the source transfer DDM approach [7, 8]. The sweeping type DDM is *multiplicative*; the subdomain problems are solved one after another in a particular order/direction, which could be interpreted as the process of LU or LDL^T factorization. The multiplicative DDMs usually have better convergence than the additive DDMs; however, the sequential sweeping order causes many difficulties to parallel computing in terms of efficiency and scalability. In addition, the subdomain problems in the DDMs are often solved with direct solvers, such as the multifrontal method [13, 28, 1] or its variant using hierarchically semiseparable structure [24, 39], but such one-direction partitions of the domain leave us subdomains of large sizes in practice and the robustness of the direct solvers is still a challenge for such subdomain problems.

The success of the sweeping type DDMs for the Helmholtz equation has demonstrated the effectiveness using the PML in the transmission conditions, and such techniques also have helped the development of additive DDMs during recent years.

Liu and Ying proposed an additive sweeping preconditioner in [29], where the computational domain is partitioned in one direction into many thin layers, and the subproblems on all respective layers are solved in parallel with direct solvers. Wei introduced an additive overlapping DDM solver in [26], which uses a modified source transfer technique. The main advantages of the additive DDMs is that they are more suitable for parallel computing and much easier to implement on massively distributed machines than multiplicative ones. In this paper, we propose, analyze, and test a new additive overlapping DDM for the Helmholtz equation (1.1) with the condition (1.2) based on the source transfer technique [7]. We focus on our illustration and analysis on two-dimensional problems with constant medium, but it is very natural to use the method as an iterative solver or a preconditioner for the variable medium problems, and it is very straightforward to extend the proposed algorithm to three-dimensional problems. In our method, all the subdomain PML problems are solved completely in parallel at each step, and the *horizontal*, *vertical*, and *corner* directional residuals on each subdomain are then passed to its corresponding neighbor subdomains as the sources for the next step. Based on the source transfer analysis, the proposed DDM method for the PML problem defined in the whole space of $\mathbb{R}^2$ is theoretically shown to produce the exact solution in a finite number of iterative steps for the constant medium case, and the number of steps is much smaller than those needed by existing sweeping type DDMs when a general two-way decomposition (i.e., in both x and y directions) of the domain is used. Since the practical computations of the PML problems only can be done in bounded truncated domains, we then slightly modify the proposed additive overlapping DDM to be useful for the truncated PML problem (which is an approximation to the whole space problem). An error estimate of the method is further derived, and in particular, our proof revises and simplifies some analysis techniques used in [7] and successfully generalizes them to the additive type of DDMs with the concurrent residual transfers in all directions.

The rest of the paper is organized as follows. In section 2, we first present the additive overlapping DDM for the PML problem in the whole space of $\mathbb{R}^2$ and show that the DDM solution is exactly the solution to the problem in the constant medium case. In section 3, a correspondingly modified DDM for bounded truncated domains is then developed for practical use and its error estimate is also derived. In section 4, various numerical experiments in two-dimensional and three-dimensional spaces are performed to verify the theoretical results and demonstrate effectiveness of the DDM method as an iterative solver or a preconditioner for the global discrete system. Some concluding remarks are finally drawn in section 5.

2. The additive overlapping DDM in $\mathbb{R}^2$. In this section, the PML method and the source transfer technique are first briefly reviewed, and then we will develop and analyze an additive overlapping DDM for solving the PML equation in the whole space of $\mathbb{R}^2$. The proposed DDM method will make use of the structured overlapping domain decomposition together with source transfers in horizontal, vertical, and corner directions. The constant medium problem (i.e., the constant wave number $k(\mathbf{x}) \equiv k$) is assumed for development and analysis of our DDM method.

2.1. Perfectly matched layer and source transfer. The solution of the Helmholtz equation (1.1) with the Sommerfeld radiation condition (1.2) in the constant medium case can be represented by

$$(2.1) \quad u(\mathbf{x}) = - \int_{\mathbb{R}^2} f(\mathbf{y}) G(\mathbf{x}, \mathbf{y}) d\mathbf{y} \quad \forall \mathbf{x} \in \mathbb{R}^2,$$

where $G(\mathbf{x}, \mathbf{y}) = \frac{i}{4} H_0^{(1)}(k|\mathbf{x}-\mathbf{y}|)$ is the fundamental solution of the Helmholtz equation defined by

$$(2.2) \quad \Delta G(\mathbf{x}, \mathbf{y}) + k^2 G(\mathbf{x}, \mathbf{y}) = -\delta_{\mathbf{y}}(\mathbf{x}) \quad \forall \mathbf{x} \in \mathbb{R}^2.$$

Note that $H_0^{(1)}(z)$ for $z \in \mathbb{C}$ is often referred as the first Hankel function of order zero [2].

Consider a rectangular box $B = \{(x_1, x_2)^T : a_j \leq x_j \leq b_j, j = 1, 2\} \in \mathbb{R}^2$, and the center of the rectangle is denoted by (c_1, c_2) , where $c_j = \frac{a_j+b_j}{2}$, $j = 1, 2$. We will build and solve the PML equation in $\mathbb{R}^2$ for the rectangular box B , for instance, replace the solution outside B by the complex coordinate stretching as done in the so-called uniaxial PML method [3, 9, 25, 5, 7]. Let $\alpha_1(x_1) = 1 + i\sigma_1(x_1)$, $\alpha_2(x_2) = 1 + i\sigma_2(x_2)$ with $\{\sigma_j\}_{j=1}^2$ being piecewise smooth functions such that

$$(2.3) \quad \sigma_j(x) = \begin{cases} \tilde{\sigma}(x_j - b_j) & \text{if } b_j \leq x_j, \\ 0 & \text{if } a_j < x_j < b_j, \\ \tilde{\sigma}(a_j - x_j) & \text{if } x_j \leq a_j, \end{cases}$$

where $\tilde{\sigma}(t)$ is a smooth medium profile function satisfying that $\tilde{\sigma}(t) = \gamma_0$ for $t > d$, and both d and γ_0 are some positive constants. Then the complex coordinate stretching $\tilde{\mathbf{x}}(\mathbf{x})$ for $\mathbf{x} = (x_1, x_2)$ is defined as

$$(2.4) \quad \tilde{x}_j(x_j) = c_j + \int_{c_j}^{x_j} \alpha_j(t) dt = x_j + i \int_{c_j}^{x_j} \sigma_j(t) dt, \quad j = 1, 2.$$

Denote $z^{1/2}$ the analytic branch of $\sqrt{z}$ such that $\text{Re}(z^{1/2}) > 0$ for $z \in \mathbb{C}[0, +\infty]$, and the distance in the complex plane is defined as

$$(2.5) \quad \rho(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = [(\tilde{x}_1(x_1) - \tilde{y}_1(y_1))^2 + (\tilde{x}_2(x_2) - \tilde{y}_2(y_2))^2]^{1/2}.$$

Now let us define

$$(2.6) \quad \tilde{u}(\mathbf{x}) = u(\tilde{\mathbf{x}}(\mathbf{x})) = - \int_{\mathbb{R}^2} f(\mathbf{y}) G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) d\mathbf{y} \quad \forall \mathbf{x} \in \mathbb{R}^2,$$

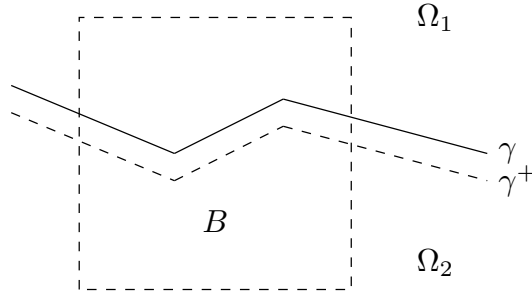
where $G(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = \frac{i}{4} H_0^{(1)}(k\rho(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}))$. We also assume that f is compactly supported in B , and then $\tilde{u}$ is well-defined in $H_{loc}^1(\mathbb{R}^2)$, satisfies $\tilde{u}(\mathbf{x}) = u(\mathbf{x})$ in B , and decays exponentially as $|x| \rightarrow \infty$. Consequently, it is the solution of the following PML equation with the source f :

$$(2.7) \quad J_B^{-1} \nabla \cdot (A_B \nabla \tilde{u}) + k^2 \tilde{u} = f \quad \forall \mathbf{x} \in \mathbb{R}^2,$$

where $A_B(x) = \text{diag}(\frac{\alpha_2(x_2)}{\alpha_1(x_1)}, \frac{\alpha_1(x_1)}{\alpha_2(x_2)})$ and $J_B(x) = \alpha_1(x_1)\alpha_2(x_2)$. We denote by $\mathcal{L}_B := J_B^{-1} \nabla \cdot (A_B \nabla \cdot) + k^2$ the linear operator associated with (2.7). The weak formulation of the PML problem (2.7) can be given as follows: for $f \in H^1(\mathbb{R}^2)'$, find $u \in H^1(\mathbb{R}^2)$ such that

$$(2.8) \quad (A_B \nabla \tilde{u}, v) - k^2 (J_B \tilde{u}, v) = -\langle J_B f, v \rangle \quad \forall v \in H^1(\mathbb{R}^2),$$

where $(\cdot, \cdot)$ denotes the inner product in $L(\mathbb{R}^2)$ and $\langle \cdot, \cdot \rangle$ the duality pairing between $H^1(\mathbb{R}^2)'$ and $H^1(\mathbb{R}^2)$. The well-posedness of the problem (2.8) has been established

FIG. 2.1. Divide $\mathbb{R}^2$ and the box B with γ .

and can be found in [7, Lemma 3.3]. Let us denote by $\mathcal{P}_B$ the above PML problem (2.8) associated with the rectangular box B .

The source transfer technique in $\mathbb{R}^2$ is briefly restated as follows. Suppose that $\mathbb{R}^2$ is divided into two parts Ω_1 and Ω_2 by the piecewise smooth curve γ , and at the same time γ also divides the rectangular box B into two parts. Let $\Omega_1^+ = \{\mathbf{x} : \rho(\mathbf{x}, \Omega_1) \leq \tilde{d}\}$, where $\tilde{d}$ is a constant, and $\gamma^+ = \partial\Omega_1^+$, as shown in Figure 2.1. There always exists a smooth function $\beta \in C^2(\mathbb{R}^2)$ such that

$$\beta|_{\Omega_1} \equiv 1, \quad \beta|_{\mathbb{R}^2 \setminus \Omega_1^+} \equiv 0, \quad 0 \leq \beta \leq 1,$$

and

$$|\nabla \beta(\mathbf{x})| < C \quad \forall \mathbf{x} \in \Omega_1^+ \setminus \Omega_1,$$

where C denotes certain generic positive constant.

Denote by χ_{Ω_2} the characteristic function of Ω_2 . It is quite straightforward to prove the following result.

LEMMA 2.1. *Suppose the support of f is in $\Omega_1 \cap B$. Let u_0 be the solution to the PML problem $\mathcal{P}_B$ with the source f (i.e., $\mathcal{L}_B u_0 = f$ in $\mathbb{R}^2$). Given $u_1 \in H^1(\mathbb{R}^2)$ such that $u_1 = u_0 \beta$ in $\mathbb{R}^2$ and let u_2 be solution to the PML problem $\mathcal{P}_B$ with the source $-(\mathcal{L}_B u_1) \chi_{\Omega_2}$ (i.e., $\mathcal{L}_B u_2 = -(\mathcal{L}_B u_1) \chi_{\Omega_2}$ in $\mathbb{R}^2$). Then we have that $u_1 + u_2 = u_0$ in $\mathbb{R}^2$ and $u_2 = 0$ in Ω_1 .*

The two domains in Figure 2.1 and Lemma 2.1 have an overlapping region $\Omega_1^+ \setminus \Omega_1$. For convenience, we shift the media profile function $\tilde{\sigma}(t)$ by $\tilde{d}$ so that

$$(2.9) \quad \tilde{\sigma}(t) = \begin{cases} 0 & \text{if } t \leq \tilde{d}, \\ \tilde{\sigma}(t - \tilde{d}) & \text{if } t > \tilde{d}, \end{cases}$$

where $\tilde{\sigma}(t)$ is the shifted medium profile. From now on, we will denote $\tilde{\sigma}(t)$ as the above shifted medium profile for simplicity, and in this way, for the PML problem $\mathcal{P}_B$, an extended region $\{\mathbf{x} : \rho(\mathbf{x}, B) \leq \tilde{d}\} \setminus B$ is always reserved as a possible overlapping region.

We will use two types of source transfers in the proposed additive overlapping DDM method for the two-dimensional problem. Let κ_1 and κ_2 be two arbitrary constants such that $a_j \leq \kappa_j \leq b_j$, $j = 1, 2$. The first type is the transfer in x or y directions, as is shown in Figure 2.2 (left), where $\gamma = \{(x_1, x_2) : x_1 = \kappa_1\}$ or $\gamma = \{(x_1, x_2) : x_2 = \kappa_2\}$. We remark that such a type of transfer is similar to the one used in STDDM [7]. The second type is the transfer in any of four corner directions,

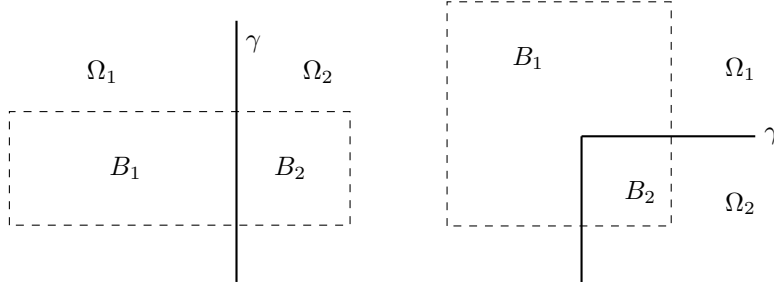


FIG. 2.2. Source transfers in the horizontal direction (left) and the corner direction (right).

as is shown in Figure 2.2 (right), where $\gamma = \{(x_1, x_2) : x_1 = \kappa_1, x_2 \leq \kappa_2\} \cup \{(x_1, x_2) : x_1 \geq \kappa_1, x_2 = \kappa_2\}$. In both cases, we have $u_2 = 0$ in Ω_1 , and the PML configuration is exactly the same in Ω_2 for the PML problem $\mathcal{P}_B$ and $\mathcal{P}_{B_2}$ ($B_2 = B \cap \Omega_2$), and therefore instead of solving the PML problem $\mathcal{P}_B$, we can turn to solve the PML problem $\mathcal{P}_{B_2}$, whose solution is zero in Ω_1 .

2.2. The DDM for the PML problem in $\mathbb{R}^2$. We now propose a new additive overlapping DDM method to solve the PML equation (2.8) associated with a rectangular domain $\Omega = (-l_1, l_1) \times (-l_2, l_2)$ in $\mathbb{R}^2$. Assume that Ω is uniformly partitioned into $N_1 \times N_2$ nonoverlapping rectangular subdomains. Let $\Delta\xi = 2l_1/N_1$, $\xi_i = -l_1 + (i-1)\Delta\xi$, $i = 1, \dots, N_1 + 1$, and $\Delta\eta = 2l_2/N_2$, $\eta_j = -l_2 + (j-1)\Delta\eta$, $j = 1, \dots, N_2 + 1$. Then we have $N_1 \times N_2$ nonoverlapping rectangular subdomains as

$$\Omega_{i,j} := [\xi_i, \xi_{i+1}] \times [\eta_j, \eta_{j+1}], \quad i = 1, 2, \dots, N_1, j = 1, 2, \dots, N_2.$$

It is clear that the PML equation associated with each rectangular subdomain $\Omega_{i,j}$ needs to be solved in the DDM method. The source f , which is assumed to be compactly supported in Ω , is decomposed to

$$f_{i,j} = f \cdot \chi_{\Omega_{i,j}}, \quad i = 1, \dots, N_1, j = 1, \dots, N_2.$$

For convenience, we also define the box $\Omega_{i_0, i_1; j_0, j_1}$ ($1 \leq i_0 \leq i_1 \leq N_1 + 1$, $1 \leq j_0 \leq j_1 \leq N_2 + 1$), which consists of a set of rectangular subdomains as follows:

$$\Omega_{i_0, i_1; j_0, j_1} := \bigcup_{\substack{i_0 \leq i \leq i_1 \\ j_0 \leq j \leq j_1}} \Omega_{i,j}.$$

Notice that the PML profile as in (2.9) makes each subdomain have an overlapping region with its neighbor subdomains, and thus we next define an overlapping domain decomposition of the two-dimensional space $\mathbb{R}^2$ as

$$\tilde{\Omega}_{i,j} := (\tilde{\xi}_i - \tilde{d}, \tilde{\xi}_{i+1} + \tilde{d}) \times (\tilde{\eta}_j - \tilde{d}, \tilde{\eta}_{j+1} + \tilde{d}), \quad i = 1, \dots, N_1, j = 1, \dots, N_2,$$

where

$$\tilde{\xi}_i = \begin{cases} -\infty, & i = 1, \\ \xi_i, & i = 2, \dots, N_1, \\ +\infty, & i = N_1 + 1, \end{cases} \quad \tilde{\eta}_j = \begin{cases} -\infty, & j = 1, \\ \eta_j, & j = 2, \dots, N_2, \\ +\infty, & j = N_2 + 1. \end{cases}$$

Some notation is introduced as follows. Let $\beta_0(t)$ be a monotone function in $C^2(\mathbb{R})$, such that $\beta_0(t) = 1$ for $t \leq 0$, $\beta_0(t) = 0$ for $t \geq 1$, and $|\beta'_0(t)| < C$ for $0 < t < 1$. Define the functions

$$\begin{aligned} \beta_{\leftarrow;i} &= \begin{cases} 1, & i = 1, \\ \beta_0(\frac{\xi_i - x_1}{d}), & i = 2, \dots, N_1 + 1, \end{cases} & \beta_{\downarrow;j} &= \begin{cases} 1, & j = 1, \\ \beta_0(\frac{\eta_j - x_2}{d}), & j = 2, \dots, N_2 + 1, \end{cases} \\ \beta_{\rightarrow;i} &= \begin{cases} \beta_0(\frac{x_1 - \xi_i}{d}) & i = 1, \dots, N_1, \\ 1, & i = N_1 + 1, \end{cases} & \beta_{\uparrow;j} &= \begin{cases} \beta_0(\frac{x_2 - \eta_j}{d}) & j = 1, \dots, N_2, \\ 1, & j = N_2 + 1, \end{cases} \end{aligned}$$

and

$$\begin{aligned} \beta_{\swarrow;i,j} &= \beta_{\leftarrow;i} \beta_{\downarrow;j}, & \beta_{\searrow;i,j} &= \beta_{\rightarrow;i} \beta_{\downarrow;j}, \\ \beta_{\nwarrow;i,j} &= \beta_{\leftarrow;i} \beta_{\uparrow;j}, & \beta_{\nearrow;i,j} &= \beta_{\rightarrow;i} \beta_{\uparrow;j}, & \beta_{0;i,j} &= \beta_{\leftarrow;i} \beta_{\rightarrow;i+1} \beta_{\downarrow;j} \beta_{\uparrow;j+1}. \end{aligned}$$

These functions will be used as β in Lemma 2.1 for different cases. Denote $\mathcal{L}_{i,j}$ as the linear operator associated with the PML problem $\mathcal{P}_{\Omega_{i,j}}$ (see (2.7) and (2.8)). Define the following characteristic functions for the half spaces and the quarter spaces in $\mathbb{R}^2$:

$$\begin{aligned} \chi_{\leftarrow;i} &:= \chi_{(-\infty, \xi_i) \times (-\infty, +\infty)}, & \chi_{\rightarrow;i} &:= \chi_{(\xi_i, +\infty) \times (-\infty, +\infty)}, \\ \chi_{\downarrow;j} &:= \chi_{(-\infty, +\infty) \times (-\infty, \eta_j)}, & \chi_{\uparrow;j} &:= \chi_{(-\infty, +\infty) \times (\eta_j, +\infty)}, \\ \chi_{\swarrow;i,j} &:= \chi_{(-\infty, \xi_i) \times (-\infty, \eta_j)}, & \chi_{\searrow;i,j} &:= \chi_{(\xi_i, +\infty) \times (-\infty, \eta_j)}, \\ \chi_{\nwarrow;i,j} &:= \chi_{(-\infty, \xi_i) \times (\eta_j, +\infty)}, & \chi_{\nearrow;i,j} &:= \chi_{(\xi_i, +\infty) \times (\eta_j, +\infty)}. \end{aligned}$$

Using the above β functions, characteristic functions, and linear operators, we are able to define the transfer operator as follows:

$$\begin{aligned} \Psi_{\leftarrow;i,j}(v) &:= -\mathcal{L}_{i-1,j}(\beta_{\leftarrow;i} v) \chi_{\leftarrow;i}, \\ \Psi_{\rightarrow;i,j}(v) &:= -\mathcal{L}_{i+1,j}(\beta_{\rightarrow;i+1} v) \chi_{\rightarrow;i+1}, \\ \Psi_{\downarrow;i,j}(v) &:= -\mathcal{L}_{i,j-1}(\beta_{\downarrow;j} v) \chi_{\downarrow;j}, \\ \Psi_{\uparrow;i,j}(v) &:= -\mathcal{L}_{i,j+1}(\beta_{\uparrow;j+1} v) \chi_{\uparrow;j+1}, \\ \Psi_{\swarrow;i,j}(v) &:= -\mathcal{L}_{i-1,j-1}(\beta_{\swarrow;i,j} v) \chi_{\swarrow;i,j}, \\ \Psi_{\searrow;i,j}(v) &:= -\mathcal{L}_{i+1,j-1}(\beta_{\searrow;i+1,j} v) \chi_{\searrow;i+1,j}, \\ \Psi_{\nwarrow;i,j}(v) &:= -\mathcal{L}_{i-1,j+1}(\beta_{\nwarrow;i,j+1} v) \chi_{\nwarrow;i,j+1}, \\ \Psi_{\nearrow;i,j}(v) &:= -\mathcal{L}_{i+1,j+1}(\beta_{\nearrow;i+1,j+1} v) \chi_{\nearrow;i+1,j+1}. \end{aligned}$$

Note that there are a total of $3^2 - 1 = 8$ directions for source transfer since each subdomain could have up to eight neighbor subdomains in the two-dimensional space.

First let us illustrate the process of source transferring in the *horizontal* or *vertical* directions, and the procedure is essentially the same as the STDDM [7] with one-way domain decomposition. We below use the horizontal transferring as the example for elaboration since the vertical transferring is similar. Assuming $f_{1,j_0} \neq 0$, the PML problem $\mathcal{P}_{\Omega_{1,N_1;j_0,j_0}}$ (see Figure 2.3 (left)) with the source f_{1,j_0} is to be solved using the DDM, and let us denote its exact solution by u . We can construct the solution u by solving the subdomain PML problems on Ω_{i,j_0} for $i = 1, 2, \dots, N_1$ by horizontal source transfers in order: first, the PML problem $\mathcal{P}_{\Omega_{1,j_0}}$ with the source f_{1,j_0} is solved and the solution is denoted as u_1 , and we have $u_1 = u$ in $\{(x_1, x_2) : x_1 \leq \xi_2 + \tilde{d}\}$ as shown in Figure 2.4(a); second, solve the PML problem $\mathcal{P}_{\Omega_{2,j_0}}$ with the source $\Psi_{\rightarrow;1,j_0}(u_1) = -\mathcal{L}_{2,j_0}(\beta_{\rightarrow;2} u_1) \chi_{\rightarrow;2}$ (the solution is denoted as u_2), and then based on

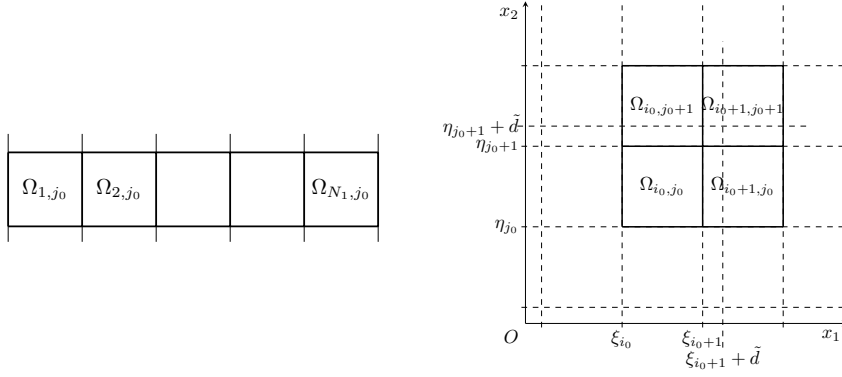


FIG. 2.3. The PML problems $\mathcal{P}_{\Omega_{1,N_1:j_0,j_0}}$ with the source f_{1,j_0} (left) and $\mathcal{P}_{\Omega_{i_0,i_0+1:j_0,j_0+1}}$ with the source f_{i_0,j_0} (right) in the DDM.

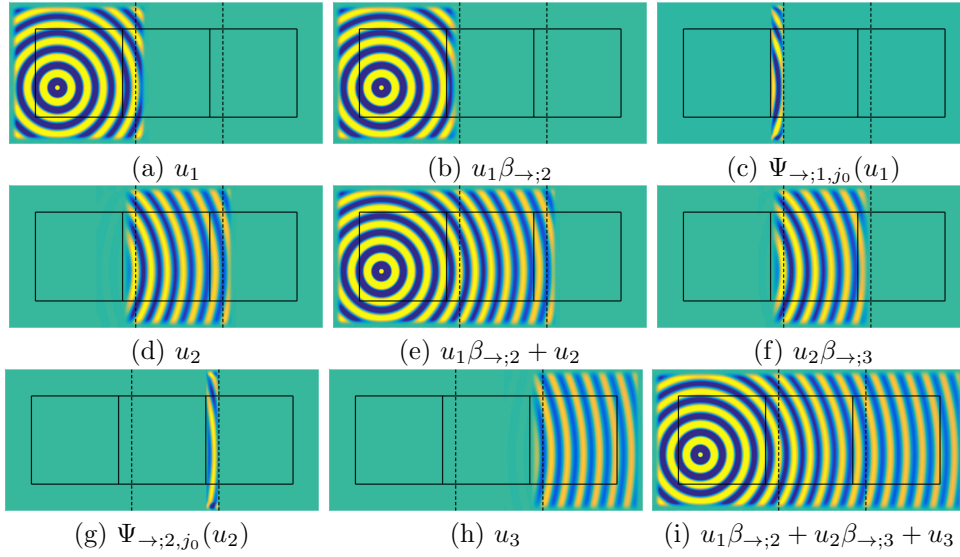


FIG. 2.4. Illustration of the process of source transferring in the right horizontal direction. The borders of subdomains Ω_{i,j_0} for $i = 1, 2, 3$ are shown with solid lines, and the right borders of overlapping regions are shown with dotted lines, which are $x_1 = \xi_2 + \tilde{d}$ and $x_1 = \xi_3 + \tilde{d}$, respectively.

Lemma 2.1 for the box $\Omega_{1,2;j_0,j_0}$, we know that $u_1\beta_{\rightarrow;2} + u_2 = u$ in $\{(x_1, x_2) : x_1 \leq \xi_3 + \tilde{d}\}$, as shown in Figure 2.4(b)–(e); third, solve the PML problem $\mathcal{P}_{\Omega_{3,j_0}}$ with the source $\Psi_{\rightarrow;2,j_0}(u_2) = -\mathcal{L}_{3,j_0}(\beta_{\rightarrow;3}u_2)\chi_{\rightarrow;3}$ (the solution is denoted as u_3), and then based on Lemma 2.1 for the box $\Omega_{1,3;j_0,j_0}$, we know that $u_1\beta_{\rightarrow;2} + u_2\beta_{\rightarrow;3} + u_3 = u$ in $\{(x_1, x_2) : x_1 \leq \xi_4 + \tilde{d}\}$, as shown in Figure 2.4(f)–(i); repeat this process until the rightmost subdomain Ω_{N_1,j_0} is reached, and finally the solution of the PML problem $\mathcal{P}_{\Omega_{1,N_1:j_0,j_0}}$ with the source f_{1,j_0} is found to be

$$u = \sum_{i=1,\dots,N_1} u_i\beta_{\rightarrow;i+1} \quad \text{in } \mathbb{R}^2.$$

It is also obvious that the PML problem $\mathcal{P}_{\Omega_{1,N_1:j_0,j_0}}$ with a source located in a middle

subdomain, e.g., $f_{i,j_0} \neq 0$ ($1 < i < N_1$), can be solved similarly with concurrent left and right source transfers.

Remark 2.2. When the source transfers are in the horizontal or vertical directions, the sources of the subdomain problems possess specific forms. Let

$$(2.10) \quad \Phi_{m;i,j}(u; \beta) := -J_{\Omega_{i,j}}^{-1} \frac{\partial}{\partial x_m} \left((A_{\Omega_{i,j}})_{m,m} \frac{\partial \beta}{\partial x_m} u \right) - J_{\Omega_{i,j}}^{-1} \frac{\partial \beta}{\partial x_m} (A_{\Omega_{i,j}})_{m,m} \frac{\partial u}{\partial x_m}$$

for $m = 1, 2$, and then the above sources for the subdomain problems could be represented, respectively, as

$$(2.11) \quad \Psi_{\rightarrow;i,j_0}(u_i) = \Phi_{1;i+1,j_0}(u_i; \beta_{\rightarrow;i+1}) \chi_{\rightarrow;i+1}, \quad i = 1, 2, \dots, N_1 - 1,$$

where the solution of other subdomain problem and its first order derivatives are involved but the wave number k is not. Similar forms exist for other horizontal or vertical directional source transfers.

Next we illustrate the process of source transferring in the corner directions of the domain decomposition. Supposing $f_{i_0,j_0} \neq 0$, the PML problem $\mathcal{P}_{\Omega_{i_0,i_0+1;j_0,j_0+1}}$ with the source f_{i_0,j_0} (see Figure 2.3 (right)) is to be solved with the DDM, and its solution is denoted as u again. First the solution to the PML problem $\mathcal{P}_{\Omega_{i_0,j_0}}$ with the source f_{i_0,j_0} is solved, and the solution is denoted as u_0 . It is obvious that

$$(2.12) \quad u_0 = u \text{ in } \{(x_1, x_2) : x_1 \leq \xi_{i_0+1} + \tilde{d} \text{ and } x_2 \leq \eta_{j_0+1} + \tilde{d}\},$$

as shown in Figure 2.5(a). Then a vertical source transfer is applied, the PML problem $\mathcal{P}_{\Omega_{i_0,j_0+1}}$ with the source $\Psi_{\uparrow;i_0,j_0}(u_0)$ is solved with the solution denoted as $u_{\uparrow}$, and we have

$$(2.13) \quad u_0 \beta_{\uparrow;j_0+1} + u_{\uparrow} = u \quad \text{in } \{(x_1, x_2) : x_1 \leq \xi_{i_0+1} + \tilde{d}\},$$

$$(2.14) \quad u_{\uparrow} = 0 \quad \text{in } \{(x_1, x_2) : x_2 \leq \eta_{j_0+1}\},$$

as shown in Figure 2.5(b)–(d). Similarly, a horizontal source transfer is applied and the problem $\mathcal{P}_{\Omega_{i_0+1,j_0}}$ with the source $\Psi_{\rightarrow;i_0,j_0}(u_0)$ is solved with the solution being denoted as $u_{\rightarrow}$, and we have

$$(2.15) \quad u_0 \beta_{\rightarrow;i_0+1} + u_{\rightarrow} = u \quad \text{in } \{(x_1, x_2) : x_2 \leq \eta_{j_0+1} + \tilde{d}\},$$

$$(2.16) \quad u_{\rightarrow} = 0 \quad \text{in } \{(x_1, x_2) : x_1 \leq \xi_{i_0+1}\},$$

as shown in Figure 2.5(e)–(g). Based on (2.12)–(2.16), we obtain that

$$(2.17) \quad u_0 \beta_{\nearrow;i_0+1,j_0+1} + u_{\rightarrow} \beta_{\uparrow;j_0+1} + u_{\uparrow} \beta_{\rightarrow;i_0+1} = u(\beta_{\rightarrow;i_0+1} + \beta_{\uparrow;j_0+1} - \beta_{\nearrow;i_0+1,j_0+1}) \text{ in } \mathbb{R}^2,$$

as shown in Figure 2.5(h)–(j). Finally the PML problem $\mathcal{P}_{\Omega_{i_0+1,j_0+1}}$ with the source

$$(2.18) \quad \begin{aligned} & -\mathcal{L}_{\Omega_{i_0,i_0+1;j_0,j_0+1}}(u(\beta_{\rightarrow;i_0+1} + \beta_{\uparrow;j_0+1} - \beta_{\nearrow;i_0+1,j_0+1})) \chi_{\nearrow;i_0+1,j_0+1}, \\ & = -\mathcal{L}_{\Omega_{i_0+1,j_0+1}}(u_0 \beta_{\nearrow;i_0+1,j_0+1} + u_{\rightarrow} \beta_{\uparrow;j_0+1} + u_{\uparrow} \beta_{\rightarrow;i_0+1}) \chi_{\nearrow;i_0+1,j_0+1} \\ & = \Psi_{\nearrow;i_0,j_0}(u_0) + \Psi_{\uparrow;i_0+1,j_0}(u_{\rightarrow}) + \Psi_{\rightarrow;i_0,j_0+1}(u_{\uparrow}) \end{aligned}$$

is solved and the solution is denoted as $u_{\nearrow}$, as shown in Figure 2.5(k). Observe that

$$\begin{aligned} & \beta_{\rightarrow;i_0+1} + \beta_{\uparrow;j_0+1} - \beta_{\nearrow;i_0+1,j_0+1} = 1 - (1 - \beta_{\rightarrow;i_0+1})(1 - \beta_{\uparrow;j_0+1}) \\ & = \begin{cases} 1 & \text{in } \{(x_1, x_2) : x_1 \leq \xi_{i_0+1} \text{ or } x_2 \leq \eta_{j_0+1}\}, \\ 0 & \text{in } \{(x_1, x_2) : x_1 \geq \xi_{i_0+1} + \tilde{d} \text{ and } x_2 \geq \eta_{j_0+1} + \tilde{d}\}. \end{cases} \end{aligned}$$

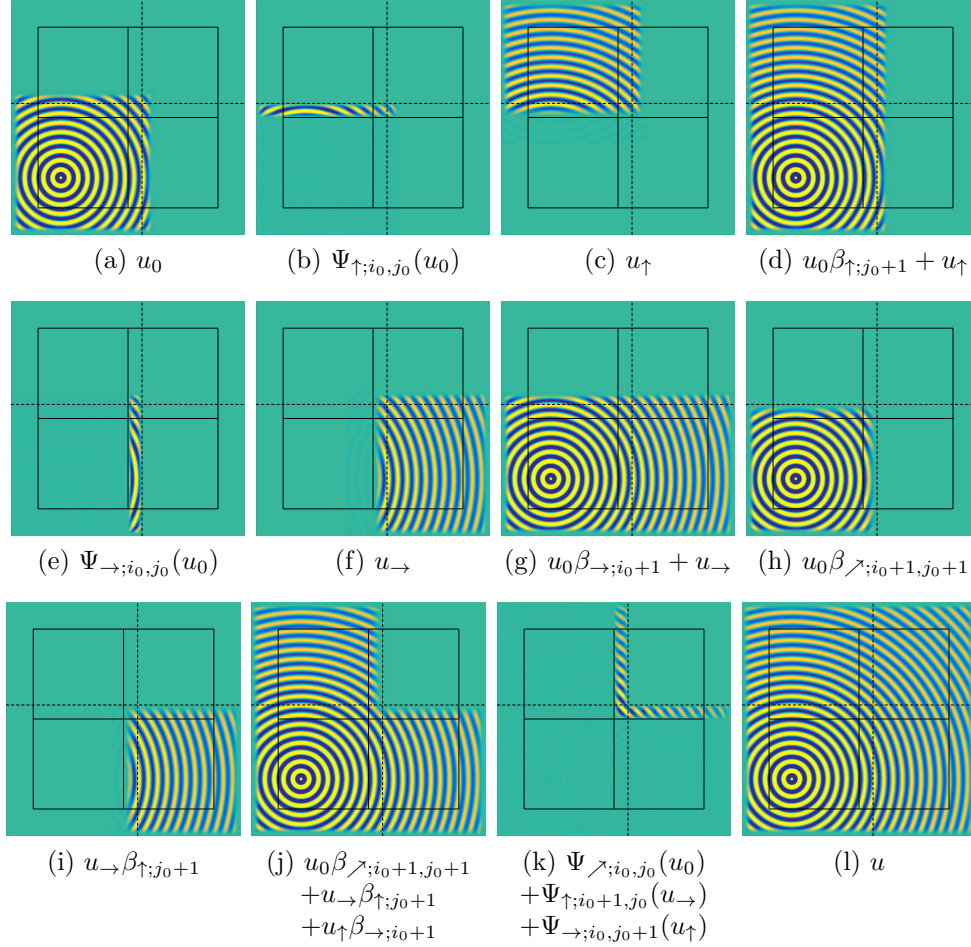


FIG. 2.5. Illustration of the process of source transferring in the upper-right corner direction. The borders of subdomains $\Omega_{i',j'}$ for $i' = i_0, i_0 + 1$ and $j' = j_0, j_0 + 1$ are shown with solid lines, and the upper and right borders of overlapping regions are shown with dotted lines, which are $x_1 = \xi_{i_0+1} + \bar{d}$ and $x_2 = \eta_{j_0+1} + \bar{d}$, respectively.

Using Lemma 2.1 on the box $\Omega_{i_0, i_0+1; j_0, j_0+1}$, the solution of the PML problem $\mathcal{P}_{\Omega_{i_0, i_0+1; j_0, j_0+1}}$ with the source f_{i_0, j_0} is finally found to be

$$u = u_0 \beta_{\nearrow; i_0+1, j_0+1} + u_{\rightarrow} \beta_{\uparrow; j_0+1} + u_{\uparrow} \beta_{\rightarrow; i_0+1} + u_{\nearrow} \quad \text{in } \mathbb{R}^2,$$

as shown in Figure 2.5(l).

Remark 2.3. When the source transfers are in the corner directions, the sources for the subdomain problems also have specific forms, e.g., the source (2.18) could be represented as

$$(2.19) \quad \left(\Phi_{1; i_0+1, j_0+1}(u_{\uparrow}; \beta_{\rightarrow; i_0+1}) + \Phi_{2; i_0+1, j_0+1}(u_{\rightarrow}; \beta_{\uparrow; j_0+1}) \right. \\ \left. + \Phi_{1; i_0+1, j_0+1}(u_0; \beta_{\nearrow; i_0+1, j_0+1}) + \Phi_{2; i_0+1, j_0+1}(u_0; \beta_{\nearrow; j_0+1, j_0+1}) \right) \chi_{\nearrow; i_0+1, j_0+1},$$

where the solutions of other subdomain problems and their first order derivatives are again involved but the wave number k is not. Similar forms exist for other corner

directional source transfers. Also such source transfers in the corner directions could be generalized to the case of more subdomains.

With the source transfers in the horizontal, vertical, and corner directions as discussed above, the solution to the PML problem $\mathcal{P}_\Omega$ with the source f_{i_0, j_0} could be constructed by induction; the details are omitted here. For the general source $f = \sum_{\substack{i=1, \dots, N_1 \\ j=1, \dots, N_2}} f_{i,j}$, the solution process for each source $f_{i,j}$ can be handled concurrently, and thus we propose an additive overlapping DDM for the PML problem as below.

ALGORITHM 2.1 (additive overlapping DDM for the PML problem $\mathcal{P}_\Omega$ with the source f).

- Set $\{u_{i,j}^0\} = 0$ in $\mathbb{R}^2$ for $i = 1, 2, \dots, N_1, j = 1, 2, \dots, N_2$.
- Step 1: solve the PML problem $\mathcal{P}_{\Omega_{i,j}}$ with the source $f_{i,j}$

$$(2.20) \quad \mathcal{L}_{i,j} u_{i,j}^1 = f_{i,j} \quad \text{in } \mathbb{R}^2$$

for $i = 1, 2, \dots, N_1, j = 1, 2, \dots, N_2$.

- Step $s = 2, 3, \dots, N_1 + N_2 - 1$: solve

$$(2.21) \quad \begin{aligned} \mathcal{L}_{i,j} u_{i,j}^s = & \Psi_{\leftarrow; i+1, j}(u_{i+1, j}^{s-1}) + \Psi_{\rightarrow; i-1, j}(u_{i-1, j}^{s-1}) \\ & + \Psi_{\downarrow; i, j+1}(u_{i, j+1}^{s-1}) + \Psi_{\uparrow; i, j-1}(u_{i, j-1}^{s-1}) \\ & + \Psi_{\swarrow; i+1, j+1}(u_{i+1, j+1}^{s-2}) + \Psi_{\searrow; i-1, j+1}(u_{i-1, j+1}^{s-2}) \\ & + \Psi_{\nwarrow; i+1, j-1}(u_{i+1, j-1}^{s-2}) + \Psi_{\nearrow; i-1, j-1}(u_{i-1, j-1}^{s-2}) \quad \text{in } \mathbb{R}^2 \end{aligned}$$

for $i = 1, 2, \dots, N_1, j = 1, 2, \dots, N_2$.

- The DDM solution for $\mathcal{P}_\Omega$ with the source f is then given by

$$(2.22) \quad u_{DDM} = \sum_{\substack{i=1, \dots, N_1 \\ j=1, \dots, N_2}} \beta_{0; i, j} u_{i, j}$$

$$\text{with } u_{i, j} = \sum_{s=1, \dots, N_1+N_2-1} u_{i, j}^s.$$

THEOREM 2.4. $u_{DDM}(f) = u(f)$, where $u(f)$ is the solution of the PML problem $\mathcal{P}_\Omega$ with the source f and $u_{DDM}(f)$ is the corresponding DDM solution constructed by Algorithm 2.1.

Proof. Obviously, $u_{DDM}(f) = \sum_{\substack{i=1, \dots, N_1 \\ j=1, \dots, N_2}} u_{DDM}(f_{i, j})$. Hence to show that $u_{DDM}(f)$ is the solution to the problem $\mathcal{P}_\Omega$ with the source f , we only need to show that for any $i_0 \in \{1, \dots, N_1\}$ and $j_0 \in \{1, \dots, N_2\}$, $u_{DDM}(f_{i_0, j_0})$ is the solution to $\mathcal{P}_\Omega$ with the source $f(i_0, j_0)$, denoted as $u(f_{i_0, j_0})$.

At Step 1, only the solution of subdomain Ω_{i_0, j_0} , u_{i_0, j_0}^1 , need to be updated with (2.20) ($u_{i, j}^1 = 0$ for $i \neq i_0$ or $j \neq j_0$ due to zero source).

At Step 2, based on (2.21), on the subdomains $\Omega_{i, j}$ with $|i - i_0| + |j - j_0| + 1 = 2$, $u_{i, j}^2$'s are computed with horizontal or vertical source transfers. For instance, on the subdomain Ω_{i_0+1, j_0} , omitting the zero terms in the right-hand side (RHS) of (2.21),

$$(2.23) \quad \mathcal{L}_{i_0+1, j_0} u_{i_0+1, j_0}^2 = \Psi_{\rightarrow; i_0, j_0}(u_{i_0, j_0}^1).$$

Using the analysis of horizontal and vertical directional source transfers, we have that

$$(2.24) \quad \sum_{\substack{r=1, 2 \\ |i-i_0|+|j-j_0|+1 \leq 2}} u_{i, j}^r \beta_{0; i, j} = u(f_{i_0, j_0}) \quad \text{in } \bigcup_{|i-i_0|+|j-j_0|+1 \leq 2} \Omega_{i, j}$$

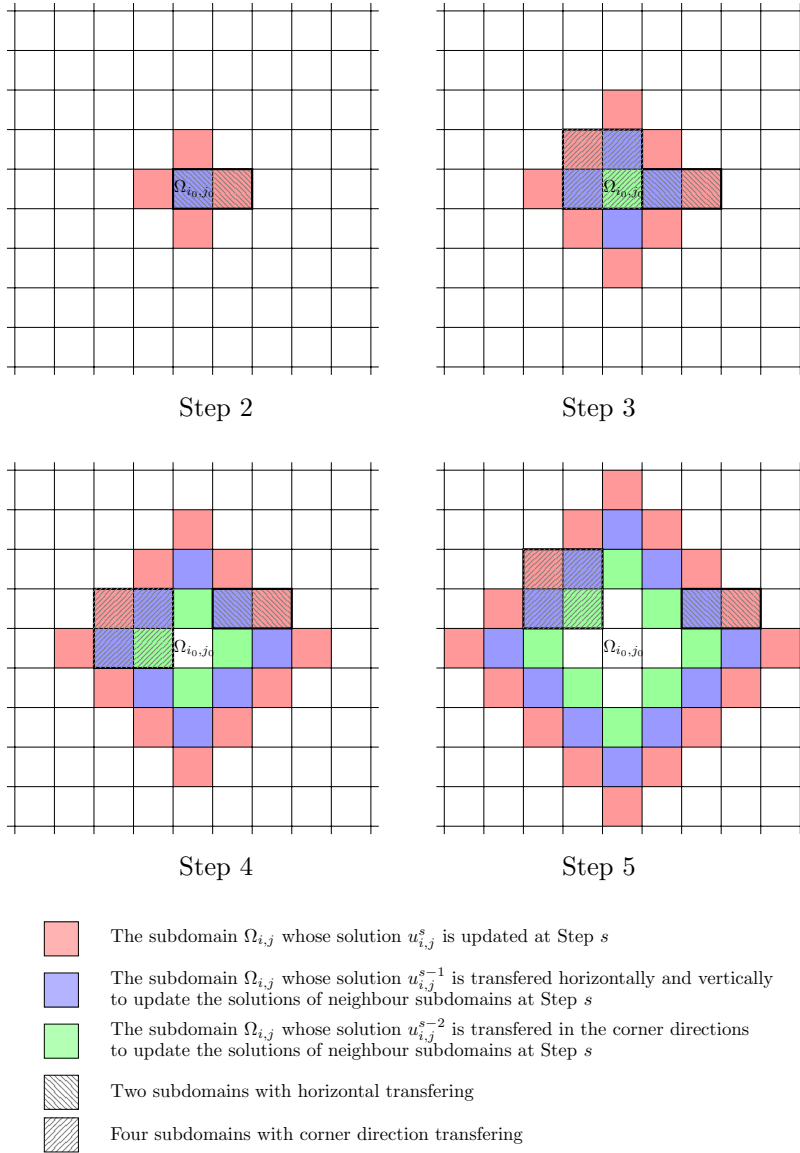


FIG. 2.6. The updating process of the subdomain solutions at Steps 2, 3, 4, 5 of Algorithm 2.1 (the additive overlapping DDM for the PML problem $\mathcal{P}_\Omega$ with the source $f(i_0, j_0)$).

as is shown in Figure 2.6 (top-left). Note that there are no reflections opposite to the propagating direction—let us take the subdomain Ω_{i_0+1, j_0} , for example, with the horizontal updating (2.23); u_{i_0+1, j_0}^2 is a right going ($\rightarrow$) wave solution, and thus by Lemma 2.1, $\Psi_{\square; i_0+1, j_0}(u_{i_0+1, j_0}^2) = 0$ for $\square = \leftarrow, \nwarrow, \swarrow$.

At Step 3, based on (2.21), on the subdomains $\Omega_{i,j}$ with $|i - i_0| + |j - j_0| + 1 \leq 3$, $u_{i,j}^3$'s are computed. However, since there are no reflections opposite to the propagating direction, only those on the subdomains $\Omega_{i,j}$ with $|i - i_0| + |j - j_0| + 1 = 3$ really need to be updated. The solutions on the subdomains $\Omega_{i_0 \pm 2, j_0}$ and $\Omega_{i_0, j_0 \pm 2}$ are updated with horizontal or vertical source transfers, respectively; the solutions on the

subdomains $\Omega_{i_0 \pm 1, j_0 \pm 1}$ are updated with horizontal, vertical, and corner directional source transfers. For instance, on the subdomain Ω_{i_0-1, j_0+1} , omitting the zero terms in the RHS of (2.21),

(2.25)

$$\mathcal{L}_{i_0-1, j_0+1} u_{i_0-1, j_0+1}^3 = \Psi_{\leftarrow; i_0, j_0+1}(u_{i_0, j_0+1}^2) + \Psi_{\uparrow; i_0-1, j_0}(u_{i_0-1, j_0}^2) + \Psi_{\nwarrow; i_0, j_0}(u_{i_0, j_0}^1).$$

Using the analysis of horizontal, vertical, and corner directional source transfers, we have that

$$(2.26) \quad \sum_{\substack{r=1, \dots, 3 \\ |i-i_0|+|j-j_0|+1 \leq 3}} u_{i,j}^r \beta_{0;i,j} = u(f_{i_0, j_0}) \quad \text{in} \quad \bigcup_{|i-i_0|+|j-j_0|+1 \leq 3} \Omega_{i,j}$$

as is shown in Figure 2.6 (top-right). Again, there are no reflections opposite to the propagating direction for the horizontal, vertical, and corner directional transfers: the case of horizontal and vertical solution updating are similar to Step 2; for the case of corner direction solution updating, let us take the subdomain Ω_{i_0-1, j_0+1} , for example, with the corner directional updating (2.25); u_{i_0-1, j_0+1}^2 is a upper-left going ($\nwarrow$) wave solution, and thus by Lemma 2.1, $\Psi_{\square; i_0-1, j_0+1}(u_{i_0-1, j_0+1}^2) = 0$ for $\square = \rightarrow, \downarrow, \nearrow, \swarrow, \searrow$.

Repeat the procedure (see the bottom panel of Figure 2.6), we always have

$$(2.27) \quad \sum_{\substack{r=1, \dots, s \\ |i-i_0|+|j-j_0|+1 \leq s}} u_{i,j}^r \beta_{0;i,j} = u(f_{i_0, j_0}) \quad \text{in} \quad \bigcup_{|i-i_0|+|j-j_0|+1 \leq s} \Omega_{i,j}.$$

Hence we finally have $u_{\text{DDM}}(f_{i_0, j_0}) = u(f_{i_0, j_0})$ after Step $N_1 + N_2 - 1$. $\square$

Remark 2.5. Algorithm 2.1 is more efficient than existing sweeping type DDMs, for example, the recursive STDDM [12] which needs a total of $N_1 N_2$ iterative steps to produce the exact solution of the PML problem.

3. The additive overlapping DDM in bounded truncated domains.

In this section, the additive overlapping DDM in a bounded truncated domain for practical computations is proposed based on the DDM in the whole space of $\mathbb{R}^2$ developed in the preceding section. Again the constant medium problem is assumed, and an error estimate of the method is established. First some notation is introduced as in [7]. For any bounded domain $U \in \mathbb{R}^2$ with Lipschitz boundary Γ , we define the following norms:

$$\|u\|_{H^1(U)} = \left(\|\nabla u\|^2 + \|ku\|_{L^2(U)}^2 \right)^{1/2}, \quad \|u\|_{H^1(U)} = \left(\|\nabla u\|^2 + d_U^{-2} \|u\|_{L^2(U)}^2 \right)^{1/2},$$

$$|v|_{\frac{1}{2}, \Gamma}^2 = \int_{\Gamma} \int_{\Gamma} \frac{|v(\mathbf{x}) - v(\mathbf{x}')|^2}{|\mathbf{x} - \mathbf{x}'|^2} ds(\mathbf{x}) ds(\mathbf{x}'), \quad \|v\|_{H^{1/2}(\Gamma)} = \left(d_U^{-1} \|\nabla v\|_{L^2(\Gamma)}^2 + |v|_{\frac{1}{2}, \Gamma}^2 \right)^{1/2},$$

where $d_U = \text{diam}(U)$. Obviously,

$$(3.1) \quad \|v\|_{H^{\frac{1}{2}}(\Gamma)} \leq (|\Gamma| d_U^{-1})^{\frac{1}{2}} \|v\|_{L^\infty(\Gamma)} + |\Gamma| \|\nabla v\|_{L^\infty(\Gamma)} \quad \forall v \in W^{1,\infty}(\Gamma).$$

Using the scaling argument and trace theorem we have for any $v \in H^{\frac{1}{2}}(\Gamma)$,

$$(3.2) \quad C_1 \frac{|U|^{\frac{1}{2}}}{|\Gamma|} \|v\|_{H^{\frac{1}{2}}(\Gamma)} \leq \inf_{\substack{\phi|_{\Gamma}=v \\ \phi \in H^1(U)}} \|\phi\|_{H^1(U)} \leq C_2 \frac{|U|^{\frac{1}{2}}}{|\Gamma|} \|v\|_{H^{\frac{1}{2}}(\Gamma)},$$

where $C_1, C_2 > 0$ are constants independent of d_U .

3.1. The DDM for the truncated PML problem. Given any rectangular domain $\Omega = [-l_1, l_1] \times [-l_2, l_2] \in \mathbb{R}^2$, let us define an extended box $\Omega^{\text{PML}} = [-l_1 - d, l_1 + d] \times [-l_2 - d, l_2 + d]$ with $d > \tilde{d}$. Then the weak formulation of the truncated PML problem associated with Ω (denoted by $\widehat{\mathcal{P}}_\Omega$) is defined by the following: for $f \in H^1(\Omega)'$, find $\hat{u} \in H_0^1(\Omega^{\text{PML}})$ such that

$$(3.3) \quad (A_\Omega \nabla \hat{u}, v) - k^2 (J_\Omega \hat{u}, v) = -\langle J_\Omega f, v \rangle \quad \forall v \in H^1(\Omega^{\text{PML}}).$$

Define $L = \max(l_1 + d, l_2 + d)$. We consider solving the above truncated PML problem $\widehat{\mathcal{P}}_\Omega$ with the source f using the additive overlapping DDM.

It is proved in [7, Lemma 3.4] that for sufficiently large $\sigma_0 d \geq 1$, where $\sigma_0 = \max_{t \in \mathbb{R}} \tilde{\sigma}(t)$, the sesquilinear form associated with the above truncated PML problem (3.3) satisfies the inf-sup condition: there exists a positive constant $\mu^{-1} \leq Ck^{\frac{3}{2}}$ such that

$$(3.4) \quad \sup_{\psi \in H_0^1(\Omega^{\text{PML}})} \frac{|(A_\Omega \nabla \phi, \nabla \psi) - k^2 (J_\Omega \phi, \psi)|}{\|\psi\|_{H^1(\Omega^{\text{PML}})}} \geq \mu \|\phi\|_{H^1(\Omega^{\text{PML}})} \quad \forall \phi \in H_0^1(\Omega^{\text{PML}}).$$

The solution of the truncated PML problem $\widehat{\mathcal{P}}_\Omega$, denoted as $\hat{u}$, is an approximation of the solution u of the PML problem $\mathcal{P}_\Omega$, and the following error estimate has been proved:

$$(3.5) \quad \|u - \hat{u}\|_{H^1(\Omega^{\text{PML}})} \leq Ck^2(1 + kL)^2 e^{-\frac{1}{2}k\gamma_0\bar{\sigma}},$$

where $\gamma_0 = \frac{d}{\sqrt{d^2 + 4((b_1 - a_1) + (b_2 - a_2) + d)^2}}$, and $\bar{\sigma} = \int_0^d \tilde{\sigma}(t) dt$. It is clear that $\hat{u}$ is zero on $\partial\Omega^{\text{PML}}$, while u is not. However, as is shown in the proof of [7, Lemma 3.4], we have the following estimate for that:

$$(3.6) \quad |u(\mathbf{x})| \leq Ck^{1/2} e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f\|_{H^1(\Omega)'} \quad \forall \mathbf{x} \in \partial\Omega^{\text{PML}},$$

$$(3.7) \quad |\nabla u(\mathbf{x})| \leq Ck^{3/2} e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f\|_{H^1(\Omega)'} \quad \forall \mathbf{x} \in \partial\Omega^{\text{PML}}.$$

The truncated PML problem that we solve in this paper is for each subdomain $\Omega_{i,j}$ ($\Omega_{i,j} \subset \Omega$), and therefore we use the same $\gamma_0 = \frac{d}{\sqrt{d^2 + 4(l_1 + l_2 + d)^2}}$ for all truncated subdomain PML problems in the rest of the paper.

Denote $\widehat{\mathcal{L}}_{i,j}$ as the operator associated with the truncated subdomain PML problem $\widehat{\mathcal{P}}_{\Omega_{i,j}}$. Substitute PML problem operator $\mathcal{L}_{i,j}$ with $\widehat{\mathcal{L}}_{i,j}$, solution $u_{i,j}$ with $\hat{u}_{i,j}$ in Algorithm 2.1, and we have the correspondingly revised DDM for the truncated PML problem in $\mathbb{R}^2$.

ALGORITHM 3.1 (additive overlapping DDM for the truncated PML problem $\widehat{\mathcal{P}}_\Omega$ with the source f).

- Set $\{\hat{u}_{i,j}^0\} = 0$ in $\Omega_{i,j}^{\text{PML}}$ for $i = 1, 2, \dots, N_1, j = 1, 2, \dots, N_2$.
- Step 1: solve the truncated PML problem $\widehat{\mathcal{P}}_{\Omega_{i,j}}$ with the source $f_{i,j}$

$$(3.8) \quad \widehat{\mathcal{L}}_{i,j} \hat{u}_{i,j}^1 = f_{i,j} \quad \text{in } \Omega_{i,j}^{\text{PML}}$$

for $i = 1, 2, \dots, N_1, j = 1, 2, \dots, N_2$.

- Step $s = 2, 3, \dots, N_1 + N_2 - 1$: solve

$$\begin{aligned}
 \widehat{\mathcal{L}}_{i,j} \hat{u}_{i,j}^s &= \Psi_{\leftarrow; i+1,j}(\hat{u}_{i+1,j}^{s-1}) + \Psi_{\rightarrow; i-1,j}(\hat{u}_{i-1,j}^{s-1}) \\
 &\quad + \Psi_{\downarrow; i,j+1}(\hat{u}_{i,j+1}^{s-1}) + \Psi_{\uparrow; i,j-1}(\hat{u}_{i,j-1}^{s-1}) \\
 &\quad + \Psi_{\swarrow; i+1,j+1}(\hat{u}_{i+1,j+1}^{s-2}) + \Psi_{\searrow; i-1,j+1}(\hat{u}_{i-1,j+1}^{s-2}) \\
 (3.9) \quad &\quad + \Psi_{\nwarrow; i+1,j-1}(\hat{u}_{i+1,j-1}^{s-2}) + \Psi_{\nearrow; i-1,j-1}(\hat{u}_{i-1,j-1}^{s-2}) \quad \text{in } \Omega_{i,j}^{PML}
 \end{aligned}$$

for $i = 1, 2, \dots, N_1$, $j = 1, 2, \dots, N_2$.

- The DDM solution for $\widehat{P}_\Omega$ with the source f is then given by

$$(3.10) \quad \hat{u}_{DDM}(f) = \sum_{\substack{i=1, \dots, N_1 \\ j=1, \dots, N_2}} \beta_{0;i,j} \hat{u}_{i,j}$$

$$\text{with } \hat{u}_{i,j} = \sum_{s=1, \dots, N_1+N_2-1} \hat{u}_{i,j}^s.$$

Remark 3.1. The additive overlapping DDM algorithms (Algorithms 2.1 and 3.1) can be straightforwardly generalized to solve the three-dimensional PML and truncated PML problems with a general three-way domain composition of $N_1 \times N_2 \times N_3$ subdomains, in which there are now a total of $3^3 - 1 = 26$ directions for source transferring and $N_1 + N_2 + N_3 - 1$ steps.

3.2. Error estimate. Let us first only consider the case of the source $f_{i_0, j_0} \neq 0$. Then it holds that the subdomain solution $u_{i,j}^s$ is nonzero only at Step $s = |i - i_0| + |j - j_0| + 1$, while the subdomain solution $\hat{u}_{i,j}^s$ is nonzero at Steps $s \geq |i - i_0| + |j - j_0| + 1$.

LEMMA 3.2. Assume only the source $f_{i_0, j_0} \neq 0$ and $\sigma_0 d$ to be sufficiently large. For $i = 1, \dots, N_1$, $j = 1, \dots, N_2$, it holds that

$$(3.11) \quad \|u_{i,j}^s\|_{H^{1/2}(\partial\Omega_{i,j}^{PML})} \leq Ck^{\frac{1}{2}}(1 + kL)e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0, j_0}\|_{H^1(\Omega)'}$$

for $s = |i - i_0| + |j - j_0| + 1$.

Proof. First we prove that

$$(3.12) \quad |u_{i,j}^s(\mathbf{x})| \leq Ck^{\frac{1}{2}}e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0, j_0}\|_{H^1(\Omega)'}, \quad \forall \mathbf{x} \in \partial\Omega_{i,j}^{PML},$$

$$(3.13) \quad |\nabla u_{i,j}^s(\mathbf{x})| \leq Ck^{\frac{3}{2}}e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0, j_0}\|_{H^1(\Omega)'}, \quad \forall \mathbf{x} \in \partial\Omega_{i,j}^{PML}.$$

Consider the case $i \geq i_0 + 1$, $j \geq j_0 + 1$, for example. Denote by $u_{i_0, i; j_0, j}$ the solution to the PML problem $\mathcal{P}_{\Omega_{i_0, i; j_0, j}}$ with f_{i_0, j_0} as the source, and then we have by (3.6)–(3.7)

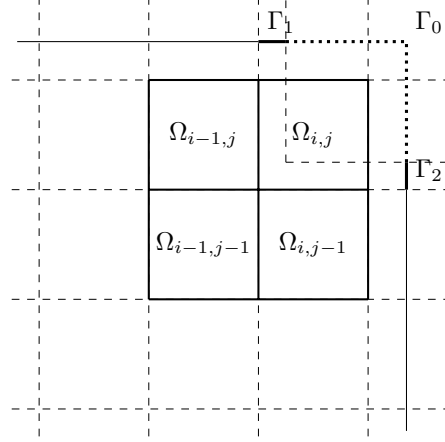
$$(3.14) \quad |u_{i_0, i; j_0, j}| \leq Ck^{\frac{1}{2}}e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0, j_0}\|_{H^1(\Omega)'}, \quad \forall \mathbf{x} \in \partial\Omega_{i_0, i; j_0, j},$$

$$(3.15) \quad |\nabla u_{i_0, i; j_0, j}| \leq Ck^{\frac{3}{2}}e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0, j_0}\|_{H^1(\Omega)'}, \quad \forall \mathbf{x} \in \partial\Omega_{i_0, i; j_0, j}.$$

Define $\Gamma_0 = \partial\Omega_{i,j}^{PML} \cap \{x_1 \geq \xi_i + d, x_2 \geq \eta_j + d\}$, $\Gamma_1 = \{\xi_i \leq x_1 \leq \xi_i + d, x_2 = \eta_{j+1} + d\}$, and $\Gamma_2 = \{x_1 = \xi_{i+1} + d, \eta_j \leq x_2 \leq \eta_{j+1} + d\}$, as shown in Figure 3.1. By Lemma 2.1, $u_{i,j}^s$ is zero on $\partial\Omega_{i_0, i; j_0, j}$ except for $\Gamma_0 \cup \Gamma_1 \cup \Gamma_2$, and moreover, by (2.17),

$$\begin{aligned}
 u_{i,j}^s &= u_{i_0, i; j_0, j} && \text{on } \Gamma_0, \\
 u_{i,j}^s &= u_{i_0, i; j_0, j}(1 - \beta_{\rightarrow; i_0}) && \text{on } \Gamma_1, \\
 u_{i,j}^s &= u_{i_0, i; j_0, j}(1 - \beta_{\uparrow; j_0}) && \text{on } \Gamma_2,
 \end{aligned}$$

which imply (3.12)–(3.13) together with (3.14)–(3.15).

FIG. 3.1. $\Gamma_0, \Gamma_1, \Gamma_2$.

By (3.1) we then obtain

$$(3.16) \quad \|u_{i,j}^s\|_{H^{1/2}(\partial\Omega_{i,j}^{\text{PML}})} \leq C \max_{x \in \partial\Omega_{i,j}^{\text{PML}}} (|u_{i,j}^s| + L|\nabla u_{i,j}^s|),$$

which further gives us (3.11) by using (3.12) and (3.13). $\square$

LEMMA 3.3. Assume only the source $f_{i_0,j_0} \neq 0$ and $\sigma_0 d$ to be sufficiently large. For $i = 1, \dots, N_1, j = 1, \dots, N_2$, it holds that

$$(3.17) \quad \|u_{i,j}^s - \hat{u}_{i,j}^s\|_{H^1(\Omega_{i,j}^{\text{PML}})} \leq C k^{\frac{3}{2}s + \frac{1}{2}} (1 + kL)^2 e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0,j_0}\|_{H^1(\Omega)}.$$

Proof. Let the source of the PML problem $\mathcal{P}_{\Omega_{i,j}}$ at Step s in (2.20)–(2.21) of the DDM Algorithm 2.1 be denoted as $f_{i,j}^s$, and the source for the truncated PML problem $\hat{\mathcal{P}}_{\Omega_{i,j}}$ at Step s in (3.8)–(3.9) of the DDM algorithm 3.1 be denoted as $\hat{f}_{i,j}^s$.

First we will prove the following error estimate for the source in the DDM Algorithm by induction:

$$(3.18) \quad \|f_{i,j}^s - \hat{f}_{i,j}^s\|_{H^1(\Omega_{i,j}^{\text{PML}})} \leq C k^{\frac{3}{2}s - 1} (1 + kL)^2 e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0,j_0}\|_{H^1(\Omega)}.$$

Obviously (3.18) holds for $s = 1$ since $f_{i,j}^1 = \hat{f}_{i,j}^1$; supposing it holds for $s = 2, \dots, t-1$, we will prove it also holds for $s = t$. By the analysis of source transfers in the horizontal, vertical, and corner directions, we know that the sources of the subdomain problems have other forms that involve the solutions of other subdomain problems and their first order derivatives but not k . This indicates

$$(3.19) \quad \begin{aligned} \|f_{i,j}^t - \hat{f}_{i,j}^t\|_{H^{-1}(\Omega_{i,j}^{\text{PML}})} &\leq \sum_{(i',j')=(i\pm 1,j),(i,j\pm 1)} \|u_{i',j'}^{t-1} - \hat{u}_{i',j'}^{t-1}\|_{H^1(\Omega_{i',j'}^{\text{PML}})} \\ &+ \sum_{(i',j')=(i\pm 1,j+1),(i\pm 1,j-1)} \|u_{i',j'}^{t-2} - \hat{u}_{i',j'}^{t-2}\|_{H^1(\Omega_{i',j'}^{\text{PML}})}. \end{aligned}$$

Taking $\|u_{i+1,j}^{t-1} - \hat{u}_{i+1,j}^{t-1}\|_{H^1(\Omega_{i+1,j}^{\text{PML}})}$, for example, by (3.2), there exists a lifting function $\phi \in H^1(\Omega_{i+1,j}^{\text{PML}})$ that $\phi = u_{i+1,j}^{t-1}$ on $\partial\Omega_{i+1,j}^{\text{PML}}$, and $\|\phi\|_{H^1(\Omega_{i+1,j}^{\text{PML}})} \leq$

$C\|u_{i+1,j}^{t-1}\|_{H^{\frac{1}{2}}(\partial\Omega_{i+1,j}^{\text{PML}})}$. Note that $u_{i+1,j}^{t-1} - \hat{u}_{i+1,j}^{t-1} - \phi \in H_0^1(\Omega_{i+1,j}^{\text{PML}})$, and then by (3.4), we have

$$\begin{aligned} & \|u_{i+1,j}^{t-1} - \hat{u}_{i+1,j}^{t-1} - \phi\|_{H^1(\Omega_{i+1,j}^{\text{PML}})} \\ & \leq \mu^{-1} \left(\sup_{\psi \in H_0^1(\Omega_{i+1,j}^{\text{PML}})} \frac{|(A_{\Omega_{i+1,j}} \nabla \phi, \nabla \psi) - k^2(J_{\Omega_{i+1,j}} \phi, \psi)|}{\|\psi\|_{H^1(\Omega_{i+1,j}^{\text{PML}})}} \right. \\ & \quad \left. + \|f_{i+1,j}^{t-1} - \hat{f}_{i+1,j}^{t-1}\|_{H^{-1}(\Omega_{i+1,j}^{\text{PML}})} \right) \\ & \leq C\mu^{-1} \left(\|\phi\|_{H^1(\Omega_{i+1,j}^{\text{PML}})} + \|f_{i+1,j}^{t-1} - \hat{f}_{i+1,j}^{t-1}\|_{H^{-1}(\Omega_{i+1,j}^{\text{PML}})} \right), \end{aligned}$$

and therefore we have

$$\begin{aligned} & \|u_{i+1,j}^{t-1} - \hat{u}_{i+1,j}^{t-1}\|_{H^1(\Omega_{i+1,j}^{\text{PML}})} \\ & \leq Ck^{\frac{3}{2}} \|f_{i+1,j}^{t-1} - \hat{f}_{i+1,j}^{t-1}\|_{H^{-1}(\Omega_{i+1,j}^{\text{PML}})} \\ & \quad + Ck^{\frac{3}{2}}(1+kL) \|\phi\|_{H^1(\Omega_{i+1,j}^{\text{PML}})} \\ & \leq Ck^{\frac{3}{2}} \|f_{i+1,j}^{t-1} - \hat{f}_{i+1,j}^{t-1}\|_{H^{-1}(\Omega_{i+1,j}^{\text{PML}})} \\ & \quad + Ck^{\frac{3}{2}}(1+kL) \|u_{i+1,j}^{t-1}\|_{H^{\frac{1}{2}}(\partial\Omega_{i+1,j}^{\text{PML}})}. \end{aligned}$$

Since $u_{i,j}^s = 0$ for $|i-i_0|+|j-j_0|+1 \neq s$, and we have (3.11) for $|i-i_0|+|j-j_0|+1 = s$ in Lemma 3.2, thus

$$\begin{aligned} & \|u_{i+1,j}^{t-1} - \hat{u}_{i+1,j}^{t-1}\|_{H^1(\Omega_{i+1,j}^{\text{PML}})} \leq Ck^{\frac{3}{2}} \|f_{i+1,j}^{t-1} - \hat{f}_{i+1,j}^{t-1}\|_{H^{-1}(\Omega_{i+1,j}^{\text{PML}})} \\ & \quad + Ck^2(1+kL)^2 e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0,j_0}\|_{H^1(\Omega)'}. \end{aligned}$$

Similar estimates also hold for the other terms in the RHS of (3.19). Therefore we have

$$\begin{aligned} & \|f_{i,j}^t - \hat{f}_{i,j}^t\|_{H^{-1}(\Omega_{i,j}^{\text{PML}})} \leq \sum_{(i',j')=(i\pm 1,j),(i,j\pm 1)} Ck^{3/2} \|f_{i',j'}^{t-1} - \hat{f}_{i',j'}^{t-1}\|_{H^1(\Omega_{i',j'}^{\text{PML}})} \\ & \quad + \sum_{(i',j')=(i\pm 1,j+1),(i\pm 1,j-1)} Ck^{3/2} \|f_{i',j'}^{t-2} - \hat{f}_{i',j'}^{t-2}\|_{H^1(\Omega_{i',j'}^{\text{PML}})} \\ & \quad + Ck^2(1+kL)^2 e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0,j_0}\|_{H^1(\Omega)'} \\ & \leq Ck^{\frac{3}{2}t-1}(1+kL)^2 e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f_{i_0,j_0}\|_{H^1(\Omega)'}, \end{aligned}$$

and thus we have (3.18) by induction.

Next we will prove (3.17). For the case $s = 1$, (3.17) holds by (3.5), and for the case $s > 1$, it also holds by using standard argument again with (3.11) and (3.18). $\square$

In the above analysis, we deal with the case of only $f_{i_0,j_0} \neq 0$. For the general case, $f = \sum_{i,j} f_{i,j}$, $u(f) = \sum_{i,j} u(f_{i,j})$, and thus we have the following result by Lemma 3.3.

THEOREM 3.4. *Assume $\sigma_0 d$ is sufficiently large, and then it holds that*

$$(3.20) \quad \|u - \hat{u}_{DDM}\|_{H^1(\Omega^{\text{PML}})} \leq Ck^{\frac{3}{2}N_1 + \frac{3}{2}N_2 + \frac{1}{2}}(1+kL)^2 e^{-\frac{1}{2}k\gamma_0\bar{\sigma}} \|f\|_{H^1(\Omega)'},$$

where u is the solution to the PML problem $\mathcal{P}_\Omega$ with the source f , and $\hat{u}_{DDM}$ is the DDM solution for the corresponding truncated PML problem constructed by Algorithm 3.1.

4. Numerical experiments. We will numerically test our additive overlapping DDM in bounded truncated domains using Algorithm 3.1. First, the convergence of the discrete DDM solutions is tested. From Theorem 3.4, the truncated DDM solution in the continuous case is a good approximation to the solution of the Helmholtz problem (1.1) when appropriate PML medium parameters are chosen. With the subdomain problems being discretized by the finite difference or finite element method, it is expected that the discretization errors usually dominate the total errors, and to verify this, we test our method for two- and three-dimensional problems with constant medium. Second, our DDM will also be tested as an iterative solver (i.e., Algorithm 3.1 could be used as an iterative method in which the total number of iterative steps is determined by certain convergence/stopping criterion) or as a preconditioner for global discrete systems for large wave number problems. We test Algorithm 3.1 for both some constant medium and layered medium problems to compare their performance. In the end we also test the performance of the proposed DDM method with the well-known Marmousi model [37], which has a complicated medium profile.

A five points finite difference scheme in two dimensions and a seven points finite difference scheme in three dimensions are, respectively, taken to discretize the problems (the Laplacian operator) on uniform rectangular meshes; both of them are obviously second order accurate. We have implemented a parallel version of DDM algorithm 3.1 using the Message Passing Interface (MPI), and the local subdomain problems are solved with the direct solver “MUMPS” [1]. The supercomputer “Tianhe-2 Cluster” located in Tianjin, China is used for our numerical tests; each node of the cluster has two 2.2GHz Xeon E5-2692 processors with 12 cores and 64G memory.

4.1. Convergence tests of the discrete DDM solutions. In this subsection, the proposed DDM is tested for the problems with constant medium—with the wave number and the number of subdomains both being fixed, the mesh resolution is increased to test the convergence of the discrete solutions.

4.1.1. Two-dimensional example. In this example, we solve a two-dimensional Helmholtz problem with a constant wave number $k/2\pi = 25$. The computational domain is $B_L = [-L, L] \times [-L, L]$ with $L = 1/2$, and the interior domain without PML is $B_l = [-l, l] \times [l, -l]$ with $l = 25/56$. Denote q as the mesh density which is the number of nodes per wave length, and a series of meshes are used in the computation where the mesh density is approximately $q \approx 22, 45, 90, 180, 270$, respectively. The domain B_L is partitioned into 5×5 subdomains, and thus the total number of steps used for constructing the DDM solution is $5 + 5 - 1 = 9$. The source is chosen as

$$f(x_1, x_2) = \frac{16k^2}{\pi^3} e^{-(\frac{4k}{\pi})^2((x_1-r_1)^2+(x_2-r_2)^2)},$$

where $(r_1, r_2) = (0.09, 0.268)$, and thus it is almost supported by four subdomains $\Omega_{3,4}$, $\Omega_{3,5}$, $\Omega_{4,4}$, and $\Omega_{4,5}$.

Table 4.1 reports the errors and convergence rates of the approximate DDM solutions for this two-dimensional example, and Figure 4.1 shows the real part of the approximate DDM solution on the mesh of 6720^2 . It is observed that the errors of the truncated DDM solutions (3.20) are very small with appropriate medium parameters for PML and the finite difference discretization errors do dominate the total errors. We obtain as expected the optimal second order convergence of the errors measured by the L^2 norm along the refinement of the meshes. Note that we also have super-convergence of the H^1 error. Such behavior was also observed in numerical tests for the Poisson problem [31] and proved on rectangular domains [35].

TABLE 4.1

The errors and convergence rates of the proposed DDM solutions for the two-dimensional constant medium problem.

Mesh size	Local size without PML	L^2 error	Conv. rate	H^1 error	Conv. rate
560^2	100^2	3.13×10^{-3}		5.47×10^{-1}	
1120^2	200^2	7.68×10^{-4}	2.0	1.36×10^{-1}	2.0
2240^2	400^2	1.95×10^{-4}	2.0	3.40×10^{-2}	2.0
4480^2	800^2	4.85×10^{-5}	2.0	8.50×10^{-3}	2.0
6720^2	1200^2	2.16×10^{-5}	2.0	3.79×10^{-3}	2.0

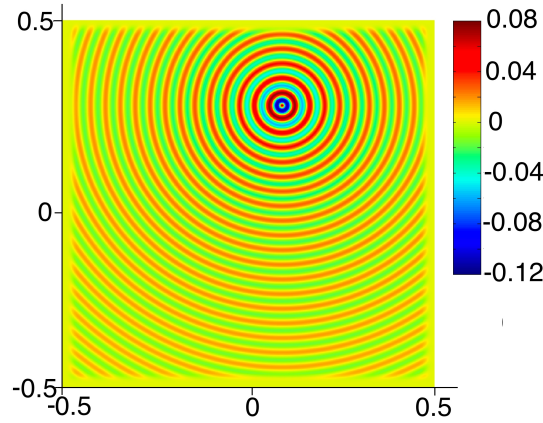


FIG. 4.1. The real part of the approximate DDM solution on the mesh of size 2240^2 for the two-dimensional constant medium problem.

4.1.2. Three-dimensional example. Next we solve a three-dimensional Helmholtz problem with a constant wave number $k/2\pi = 10$. The computational domain is still B_L with $L = 1/2$, but the interior domain without PML is B_l with $l = 3/8$. A series of meshes is used in the computation where the mesh density is approximately $q \approx 8, 9.5, 13, 16$, respectively. The domain B_L is partitioned into $3 \times 3 \times 3$ subdomains, and the total number of steps used for constructing the DDM solutions is $3 + 3 + 3 - 1 = 8$. The source is chosen as

$$f(x_1, x_2, x_3) = \frac{64k^3}{\pi^{9/2}} e^{-\left(\frac{4k}{\pi}\right)^2((x_1-r_1)^2+(x_2-r_2)^2+(x_3-r_3)^2)},$$

where $(r_1, r_2, r_3) = (0.12, 0.133, 0.125)$, and thus it is almost supported by eight subdomains $\Omega_{i,j,m}$, where $i = 2, 3$, $j = 2, 3$, $m = 2, 3$.

Table 4.2 reports the errors and convergence rates of the approximate DDM solutions for this three-dimensional example, and Figure 4.2 shows the real part of the approximate solution on the mesh of 160^3 . Again, the finite difference errors dominate the total errors, and we obtain the optimal second order convergence for both L^2 and H^1 errors as expected.

4.2. Performance tests with the DDM as an iterative solver or a preconditioner for the global discrete system. The DDM Algorithm 3.1 designed for the constant medium Helmholtz problem also could be used as an iterative solution method for both constant and variable medium problems. More steps than

TABLE 4.2

The errors and convergence rates of the proposed DDM solutions for the three-dimensional constant medium problem.

Mesh size	Local size without PML	L^2 errors	Conv. rate	H^1 errors	Conv. rate
80^3	20^3	2.66×10^{-2}		1.82×10^0	
96^3	24^3	1.81×10^{-2}	2.1	1.25×10^0	2.1
128^3	32^3	9.99×10^{-3}	2.1	6.94×10^{-1}	2.0
160^3	40^3	6.39×10^{-3}	2.0	4.45×10^{-1}	2.0

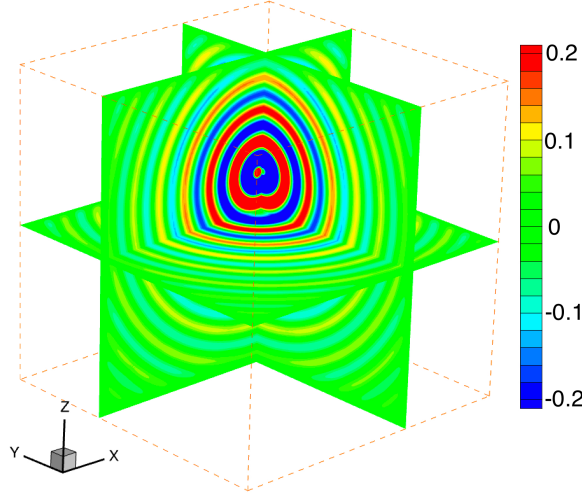


FIG. 4.2. The real part of the approximate DDM solution on the mesh of size 160^3 for the three-dimensional constant medium problem.

$N_1 + N_2 - 1$ are then usually needed for the iterative solver to reach certain relative error tolerances. We first test the DDM method as an iterative solver for some two-dimensional constant medium and layered medium problems, then for the Marmousi model problem, and at last we test the DDM method as a preconditioner for the global discrete system and compare their performance.

A single shot in the subdomain $\Omega_{1,1}$ is taken as the source with the position being (x_1^s, x_2^s) , where

$$(4.1) \quad x_1^s = \Delta\xi/4, \quad x_2^s = -\Delta\eta/3.$$

Let h_x and h_y denote the mesh sizes in x and y directions, respectively. The shape of the shot is an approximate δ function, defined by

$$(4.2) \quad f_{i,j}(x_1, x_2) = \frac{1}{h_1 h_2} \delta(x_1^s - i h_2) \delta(x_2^s - j h_2),$$

where h_1 and h_2 are the grid spacing in x and y direction, respectively.

The size of the subdomain problems is fixed in all of the following tests. Each nonoverlapping subregion is of size 300×300 , and at each side of the subdomain, the overlapping region is of 5 points width while the PML layer is of 25 points width. The number of subdomains is increased from 2×2 to 32×32 . Denote by $n_{\text{DDM-Itc}}$

TABLE 4.3

The performance of the proposed DDM as an iterative solver for the two-dimensional constant medium problem with the subdomain problem size being fixed.

Mesh size	$N_1 \times N_2$	Freq. $\omega/2\pi$	$n_{\text{DDM_Ite}}$	$n_{\text{DDM_Sol}}$	Total time (sec)	Time/step (sec)
600^2	2×2	55	5	1.67	2.0	0.40
1200^2	4×4	105	13	1.86	4.7	0.36
2400^2	8×8	205	29	1.93	11.1	0.38
4800^2	16×16	405	61	1.97	24.4	0.40
9600^2	32×32	805	123	1.95	55.9	0.45
19200^2	64×64	1605	247	1.94	117.2	0.47

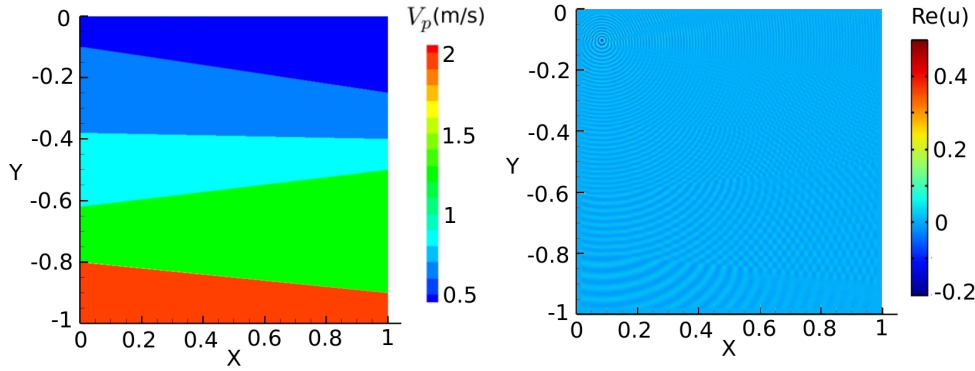


FIG. 4.3. Left: the two-dimensional layered medium velocity profile; right: the real part of the approximate DDM solution of the problem on the mesh of size 2400^2 .

the total number of iterative steps. The number of DDM solves defined by

$$(4.3) \quad n_{\text{DDM_Sol}} = \frac{n_{\text{DDM_Ite}}}{N_1 + N_2 - 1}$$

is used to measure the effectiveness of the iterative solver. It is always highly desired that the number of needed steps $n_{\text{DDM_Ite}}$ for convergence is proportional to $N_1 + N_2 - 1$ as the number of subdomains grows; in that case, $n_{\text{DDM_Sol}}$ remains almost the same. The tolerance for relative residuals is set to be 10^{-8} as the stopping criterion for all tests. Note that we also let each subdomain use exactly one core in the following tests.

4.2.1. As an iterative solver. The Algorithm 3.1 is first tested as an iterative solver for the global discrete system.

First we test the additive overlapping DDM solver for a constant medium problem on the square domain $[0, 1] \times [-1, 0]$, with different frequencies. The results on the numbers of used iterative steps and the running times are reported in Table 4.3. As we can see, the number of DDM solves $n_{\text{DDM_Sol}}$ grows as the number of the subdomains grows from 2×2 to 8×8 , but remains stable and slightly below 2 when the number of the subdomains grows further.

Next we test the DDM solver for a layered medium problem on the square domain $[0, 1] \times [-1, 0]$, as shown in Figure 4.3 (left) for the velocity profile. For the purpose of illustration, the approximate solution of the problem produced by using the uniform mesh of size 1600^2 is shown in Figure 4.3 (right). The results on the numbers of used iterative steps and the running times are reported in Table 4.4. We would like to note

TABLE 4.4

The performance of the proposed DDM as an iterative solver for the two-dimensional layered medium problem with the subdomain problem size being fixed.

Mesh size	$N_1 \times N_2$	Freq. $\omega/2\pi$	$n_{\text{DDM_Ite}}$	$n_{\text{DDM_Sol}}$	Total time (sec)	Time/step (sec)
600^2	2×2	27.5	16	5.33	5.7	0.36
1200^2	4×4	52.5	38	5.43	13.2	0.35
2400^2	8×8	102.5	73	4.87	27.8	0.38
4800^2	16×16	202.5	146	4.71	62.4	0.43
9600^2	32×32	402.5	291	4.62	133.1	0.46
19200^2	64×64	802.5	582	4.58	278.8	0.48

TABLE 4.5

The performance of the proposed DDM as a preconditioner to the global FGMRES solver for the two-dimensional constant medium problem. The number of subdomains is 16×16 .

K	$n_{\text{GMRES_Ite}}$	$n_{\text{Local_Sol}}$	Total time (sec)
1	91	91	44.0
2	41	82	39.8
3	27	81	37.8
5	16	80	35.0
10	8	80	33.3
15	5	75	30.7
20	4	80	32.9
32	2	64	26.6

TABLE 4.6

The performance of the proposed DDM as a preconditioner to the global FGMRES solver for the two-dimensional layered medium problem. The number of subdomains is 16×16 .

K	$n_{\text{GMRES_Ite}}$	$n_{\text{Local_Sol}}$	Total time (sec)
1	148	148	83.7
2	73	146	71.3
3	50	150	68.5
5	30	150	66.1
10	15	150	62.3
15	10	150	60.8
20	8	160	66.9
32	5	160	63.1

that the optimized Schwarz method with overlap proposed by Gander and Zhang also shows a similar growth of the iteration numbers [22]. As we can see, the number of DDM solves almost does not grow as the number of the subdomains grows, and it requires about 2.3 times the number of iterative steps to converge compared with the constant medium problem.

Based on the results on the time cost per DDM iterative step with up to $64 \times 64 = 4096$ (equivalent to the number of subdomains) reported in Tables 4.3 and 4.4, it is easy to see that the proposed additive overlapping DDM achieve excellent parallel efficiency.

4.2.2. As a preconditioner. Next we use Algorithm 3.1 as a preconditioner to the flexible GMRES (FGMRES) solver for the global discrete system. The number of subdomains is set to 16×16 as an example. Note that when using Algorithm 3.1 as a preconditioner, the number of DDM steps K in each preconditioning solve is to be chosen, we test different K 's for both constant medium and layered medium cases, and the results are reported in Tables 4.5 and 4.6. Since the major computational

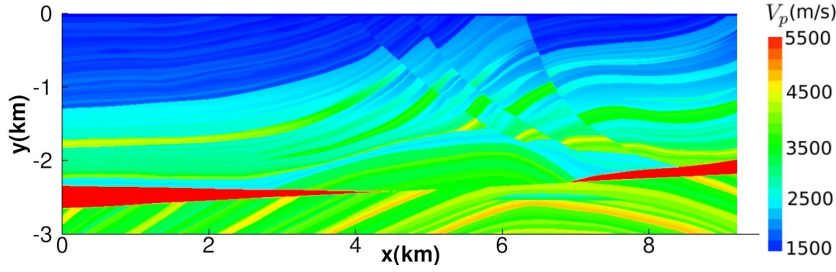


FIG. 4.4. The velocity profile of the Marmousi model.

cost of solving the linear system during each step is in the back substitute of LU factorization of each subdomain, the number of back substitutes, denoted as $n_{\text{Local_Sol}}$, is also presented in the tables.

In the constant medium case, both the total running time and $n_{\text{Local_Sol}}$ basically decrease as K increases from 1 to $N_1 + N_2 = 16 + 16 = 32$. It is also easy to see that $n_{\text{Local_Sol}}$ for all cases of K are larger than $n_{\text{DDM_Ite}} (= 61)$ used by the DDM iterative solver with 16×16 subdomains (see Table 4.3), and so does the total time cost (24.4 seconds for the DDM iterative solver). The DDM preconditioner with the largest $K = 32$ achieves almost similar performance as the DDM iterative solver.

In the layered medium case, $n_{\text{Local_Sol}}$ only slightly increases (one exception) from 148 to 160 as K increases. When $K = 1$, although $n_{\text{Local_Sol}}$ is relatively small, the extra cost of GMRES takes more time and the total running time is the largest. When $K = 2$, the best $n_{\text{Local_Sol}}$ ($= 146$) is reached and in fact it equals exactly, $n_{\text{DDM_Ite}}$, the number of iterative steps needed by the DDM iterative solver with 16×16 subdomains (see Table 4.4), but the total running time is larger (71.3 vs. 62.4 seconds). For $K \geq 3$, all the time costs are close to 62.4 seconds, which indicates that the performance differences between the DDM preconditioner and the DDM iterative solver are small.

To summarize, it is observed by our experiments that when the additive overlapping DDM is used as a preconditioner, large K should be taken, and its performance is slightly worse than being used as an iterative solver.

4.3. The Marmousi model. At last we test the performance of the proposed DDM iterative solver with the well-known Marmousi problem [37], which is a synthetic two-dimensional acoustic model from reflection seismology and is used to demonstrate the limitations of first-arrival traveltimes in imaging very complex media. The domain is $[0, 9200] \times [0, -3000]$ measured in meters; the velocity varies from 1500 m/s to 5500 m/s with the velocity profile being defined discretely through a mesh of size 2301×751 as shown in Figure 4.4. A δ source is used at the position $(\frac{N_1+1}{2}\Delta\xi, -\frac{1}{4}\Delta\eta)$. Performance results by Algorithm 3.1 with the mesh sizes and the number of subdomains being proportionally increased are reported in Table 4.7. As we can see from Table 4.7, the number of DDM solves first increases from 4 to 5.93 as the number of subdomains grows from 3×1 to 12×4 , i.e., the resolution of the mesh gradually gets better and passes the level of the velocity profile (size 2301×751). Then it remains stable and slightly under 6 as the number of the subdomains further grows. The approximate solution of the Marmousi problem obtained on the mesh of size 3600×1200 is also presented in Figure 4.5 for the purpose of visual illustration, which matches well with the benchmark results in the literature.

TABLE 4.7

The performance of the proposed DDM as an iterative solver for the Marmousi model with the subdomain problem size being fixed.

Mesh size	$N_1 \times N_2$	Freq. $\omega/2\pi$	$n_{\text{DDM_Ite}}$	$n_{\text{DDM_Sol}}$	Total time (sec)	Time/step (sec)
900×300	3×1	13	16	4.00	4.3	0.36
1800×600	6×2	25	38	4.71	13.5	0.41
3600×1200	12×4	50	89	5.93	41.9	0.47
7200×2400	24×8	98	146	5.71	82.6	0.47
14400×4800	48×16	196	291	5.22	152.8	0.46
28800×9600	96×32	392	582	5.99	365.9	0.48

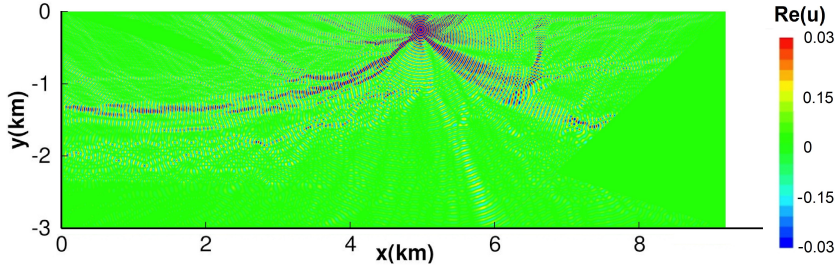


FIG. 4.5. The real part of the approximate DDM solution of the Marmousi model on the mesh of size 3600×1200 .

5. Conclusions. In this paper, we propose and analyze an additive overlapping DDM for solving the Helmholtz equation. The source transfer technique is used in the horizontal, vertical, and corner directions, to make the wave propagate away from the source subdomain by subdomain. For the constant medium case, the proposed DDM method is shown to exactly solve the PML problem defined in the whole space of $\mathbb{R}^2$, and an error estimation is established for the method used for solving the PML problem in bounded truncated domains. This method could be used as an iterative solver or a preconditioner for the global discrete system in both constant or variable medium problems, and as an iterative solver is more preferable. This DDM method is clearly very suitable for large-scale parallel computing as demonstrated by various numerical experiments. Although this method has been naturally extended to three dimensions and numerically tested, error estimation of the method for the truncated PML problem in bounded three-dimensional domains is still an open question and worthy of further study. In addition, we would like to remark that this DDM method also could be generalized to irregular domains with special domain decompositions that require each subdomain to be a convex polygon. In that case, the source transfer would be along the outward normal directions of subdomain boundaries, and convergence analysis and performance of such a generalized method need to be investigated.

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